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RESEARCH

# Ethical perspectives on AI Agents and Agentic AI

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**Abstract**
Recent developments in Artificial Intelligence (AI) have given rise to AI Agents and Agentic AI. These systems are not only capable of performing narrowly defined tasks, but can also take on more complex activities with minimal to no human involvement. While there is already some technical research on these systems, their ethical evaluation has so far received little attention. This paper provides an initial perspective on how recent developments in AI Agents and Agentic AI relate to classical ethical principles in the context of AI, focusing on *transparency*, *justice and fairness*, *non-maleficence*, *responsibility and accountability*, and *privacy*. To enable responsible development and deployment, we highlight three particularly crucial aspects of these systems: (1) their broad agency which (2) has a higher impact on human autonomy, leading to (3) an even deeper entanglement between humans and technology compared to traditional AI systems. Our analysis, which uses the method of the *wide reflective equilibrium*, serves as a starting point for ethical reflection and underscores the urgent need for ongoing interdisciplinary engagement to ensure that these novel AI systems are developed and used responsibly.

**Keywords** Ethics · Artificial intelligence · Governance · Emerging technologies

## 1 Introduction

Artificial Intelligence (AI) has become a widely discussed topic and is increasingly applied across diverse domains of society [1–3]. Recent trends and developments suggest a new level of capability under the emerging concepts of *AI Agents* and *Agentic AI* [4, 5]. While AI Agents (i.e., a *single-agent architecture*) are capable of performing specific, more narrowly defined tasks (e.g., a Chatbot in customer service connected to a knowledge base), the interconnection of multiple AI Agents in *Agentic AI* systems (i.e., a *multi-agent structure*) is designed to take on broader, more complex responsibilities, sometimes functioning as personal assistants capable of performing a wide range of everyday tasks. Acting with relative technical autonomy [6], they are envisioned to pursue complex goals with minimal to no human intervention.

For instance, one can imagine a scenario in which an AI system independently purchases concert tickets, books a train to the venue, reserves a hotel, and informs friends that all arrangements are complete. Or an AI system in the field of medicine that autonomously coordinates medical appointments according to the patient's physical states. Such cases illustrate: while AI Agents and Agentic AI systems are powerful in solving complex tasks, they require almost unrestricted access to sensitive personal data, such as names, birth dates, health data, credit card information, and even private communications, which already touches on the ethical implications of these systems, such as those related to privacy and transparency. Their capacity to act with a high degree of technical autonomy and with minimal to no human intervention underscores both their enormous potential and their ambivalence and raises questions regarding responsibility, fairness and non-maleficence. On the one hand, they hold the promise of relieving everyday burdens by taking over tedious administrative or repetitive tasks and thereby saving time. On the other hand, the level

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of autonomy and extensive data access highlight significant ethical concerns.

Despite growing *technical* research on AI Agents and Agentic AI, *ethical* engagement with these developments—despite first considerations [7, 8]—remains largely underdeveloped. Their impact on human autonomy, broad agency as well as comprehensive access to personal information amplify classical ethical questions in AI, including transparency, fairness and responsibility. In this paper, we build on the principles from UNESCO's [9] Recommendation on the Ethics of Artificial Intelligence in a structure based on the most frequently discussed principles presented by Jobin et al. [10] to examine how AI Agents and Agentic AI relate to existing ethical principles. The purpose of this work is therefore to provide an initial ethical perspective on these emerging systems, highlighting potential risks and challenges, and to encourage further discourse and research. In this way, it also contributes to the normative reflection of these novel systems, aiming to ethically guide their responsible development and deployment.

## 2 AI Agents and Agentic AI

Many AI systems have been developed for specific contexts. For example, an AI system in radiology may assist with interpreting imaging data, while another in industry may identify defective components. These narrowly defined applications are now being expanded by AI Agents and Agentic AI.

In their paper, Sapkota et al. [5] conceptually distinguish between the terms *AI Agent* and *Agentic AI*, providing a clearer distinction for understanding the latest developments in AI research that is commonly found [11–13]. According to Sapkota et al. [5], "AI Agents are typically designed as single-entity systems that perform goal-directed tasks by utilizing external tools, applying sequential reasoning, and integrating real-time information to complete well-defined functions". An AI Agent—structurally configured as a *single agent architecture* [14]—is therefore characterized by a certain level of agency (understood in a gradual and multidimensional sense following Misselhorn [15], where basic agency presupposes only intelligence and autonomy in a weak sense (not in a personal or moral manner), thus emphasizing the intrinsic dynamics of AI systems and allowing them to be perceived as capable of action [16], thereby reframing questions of responsibility), task-specificity, as well as reactivity and adaptability. Unlike classical generative AI and large language models (LLMs), an AI Agent is not limited to producing text, images, or videos based on training data, rather, it can access external information through APIs and operate tools. Examples of such AI Agents include LLMs connected to knowledge bases for customer support, or autonomous scheduling assistants [5]. These systems can assist with specific tasks, e.g. in customer support an LLM-based chatbot is connected to databases or ticketing systems and can give answers based on that information or a scheduling assistant can automatically arrange appointments in a calendar. However, their scope of action remains relatively well-defined, and their level of agency is still limited.

Beyond this *single agent architecture*, there is the *multi agent architecture*, which Sapkota et al. [5] categorize under the term *Agentic AI*. These systems "are composed of multiple, specialized agents that coordinate, communicate, and dynamically allocate sub-tasks within a broader workflow to achieve a common goal(s)". In other words, multiple AI Agents are combined to pursue more complex objectives. According to Acharya et al. [4], "Agentic AIs constitute a qualitative leap in the development of artificial intelligence, defined by their capability to set complex goals in a changing and uncontrolled situation and to pursue them through autonomously managing their resources". This description highlights at least three new dimensions and capabilities of Agentic AI in comparison with a single agent AI system: increased complexity, enhanced adaptability, and a heightened level of agency, which, within this complex network of individual agents, raises—among others—questions of transparency and responsibility anew and with heightened urgency.

First, increased complexity means that Agentic AI systems are no longer confined to a single specific task. Instead, they are interconnected and capable of assisting in more complex scenarios. Second, enhanced adaptability allows these systems to adjust to novel contexts more effectively. While individual AI Agents can already adapt to a certain degree, the combination of multiple agents within Agentic AI allows for even more adaptability and dynamism. Third, Agentic AIs exhibit heightened agency. While AI Agents could already act with relative technical autonomy within their narrowly defined tasks, Agentic AI extends this even further. Instead of relying on various different AI tools for specific use cases, users can delegate tasks to an Agentic AI system, which independently selects the appropriate tools and executes actions with minimal to no human intervention. Together, these three capabilities—higher complexity, adaptability, and agency—enable Agentic AI systems to assist across an even broader range of tasks and domains.

It is clear that, although AI Agents and Agentic AI can be conceptually distinguished and differ in scope and sophistication, they nonetheless share certain characteristics: both act in a goal-directed manner, operate with relative autonomy, select the tools they require, and interact with their environment to varying degrees. As a result, these new

capabilities and their effects on human-technology relations can be described along at least three dimensions, as will become evident at various points throughout our paper: It is (1) the broad agency of AI Agents and Agentic AI systems, which (2) leads to an increased impact on human autonomy, and (3) results in an even deeper entanglement between humans and technology compared to traditional AI systems. This expansion of capabilities and scope raises pressing ethical questions, which we will highlight in the following section.

## 3 Ethical perspectives

Novel AI systems in the form of AI Agents and Agentic AI are characterized mainly by broad agency, a heightened influence on human autonomy, and an increased degree of entanglement between humans and technology compared to traditional AI systems. These characteristics imply that both the positive potential and the associated risks of these systems are amplified compared to those of traditional AI, which typically operate within very narrowly defined domains and with more human intervention. It is therefore essential to examine these new systems through the lens of established ethical principles in AI. The following subsections outline several key ethical issues that arise in this context.

As a starting point, we draw on the principles in the Recommendation on the Ethics of Artificial Intelligence by the UNESCO [9]. In this document, the UNESCO defined the values and principles that "should be respected by all actors in the AI system life cycle, in the first place and, where needed and appropriate, be promoted through amendments to the existing and elaboration of new legislation, regulations and business guidelines". These principles are grounded in several core values, such as human rights and human dignity, the flourishing of the environment and ecosystem, diversity and inclusiveness as well as peace and justice. To make these underlying values more operational and actionable, UNESCO condensed them into the following ten principles: (1) proportionality and do no harm, (2) safety and security, (3) fairness and non-discrimination, (4) sustainability, (5) right to privacy, and data protection, (6) human oversight and determination, (7) transparency and explainability, (8) responsibility and accountability, (9) awareness and literacy, (10) multi-stakeholder and adaptive governance and collaboration.

To enhance clarity, we clustered these ten principles into a structure of five core principles, drawing on Jobin et al. [10], who identified these as the ones most frequently discussed within the AI ethics discourse: (1) *transparency*, (2) *justice and fairness*, (3) *non-maleficence*, (4) *responsibility* and (5) *privacy*. Building on these principles—and integrating the remaining UNESCO principles into related categories—we examine how they still apply and may be challenged by the emergence of AI Agents and Agentic AI. This paper aims to identify potential areas of concern and to serve as a starting point for deeper ethical reflection on these new and amplified challenges posed by these systems. Methodologically, this work aligns with a specific approach of applied ethics (*concrete ethics*), drawing on the *wide reflective equilibrium* as its guiding method [17, 18]. Within this evaluative approach, concrete problem constellations, moral judgments and intuitions, ethical principles and criteria, legal frameworks, as well as further background information are brought into critical dialogue with one another. The aim is to arrive at a form of equilibrium solution in which mediation between differing interests, values, and moral conceptions is achieved. The outcome of this process can then be productively utilized by being translated into governance structures.

### 3.1 Transparency

Transparency is one of the most frequently emphasized principles in AI guidelines. However, transparency is a multifaceted concept, and its meaning varies across disciplines. Drawing on Felzmann et al. [19], transparency can be understood along three dimensions: as a virtue, as a relation and as a system. In this sense, transparency entails that an AI system discloses its "operations, behavior, intentions, or considerations", always in relation to those who must interpret and act upon this openness, and always within a specific context or system. The demand for transparency in AI systems—often framed under the notion of *Explainable AI*—has proven to be highly challenging in practice, a difficulty commonly discussed under the label of the "black box" problem [20, 21].

For AI Agents, the problem of transparency becomes even more complex from at least three perspectives: (1) the AI system itself, (2) the depth and breadth of its intervention, and (3) its recipients.

First, with regard to the AI system itself, transparency becomes increasingly difficult when the system carries out complex, multi-step operations with the aid of external tools, or in the case of Agentic AI, when it is composed of multiple interacting agents. By breaking tasks down into smaller steps that are executed by different specialised systems across several stages, the system as a whole becomes even more complex and less comprehensible. This has implications for the question of human oversight and the possibilities for awareness and literacy. For if complete opacity prevails, how can humans, in practice, meaningfully intervene in such systems at all? As AI Agents and Agentic AI systems possess a

higher degree of agency, systemic interconnection and technical autonomy, the issue of whether a human remains *in the loop*, *on the loop*, or *out of the loop* becomes even more pressing compared to traditional AI systems. This question concerning the role of humans excludes—at the very least at the level of control—any approach that places humans out of the loop. However, how exactly the human role should be shaped, depends on the individual case and the specific context. Therefore, in principle, a risk-based approach, such as the one introduced in the EU AI Act, appears reasonable. On a conceptual level, the key concern is the remaining of the *human in command* [22] or that *meaningful human control* [23] is ensured. This includes, *inter alia*, participatory design, situation awareness, as well as supervisory control and risk analysis and management [22]. It thereby becomes evident that the more agency and technical autonomy these systems possess, the more participation—especially by those affected—becomes crucial [24].

Second, the use of APIs and external tools—through which the system interacts with a variety of other systems—further increases both the depth and breadth of its intervention. To provide meaningful support, an AI Agent or Agentic AI system must have extensive access to data and tools, operate with relative technical autonomy, and act on behalf of the user. This, however, makes it increasingly difficult to trace which agents communicate with one another, which systems contributed to the task, where particular information originated, and, in the case of an error, where the source of the problem lies (in an agent's algorithm, in the interaction between agents, in training data, or in externally connected data sources). A further question concerns the user's knowledge of which AI Agents, developed by which companies, located in which countries, and under which jurisdiction were involved in completing the task. The fact that many AI Agents and Agentic AI systems are developed by globally distributed private companies has significant implications for data sovereignty, both at the individual level and at the level of states.

Third, ethical concerns also emerge from the perspective of the recipients interacting with these systems. Apple's call screening feature provides an illustrative example: a virtual agent intercepts incoming calls to filter out spam. The agent engages with the caller, asking for their name and purpose of the call, before informing them that it will check whether the intended recipient is available. This raises an important question, especially given the higher technical autonomy and agency of such systems compared to earlier AI technologies: in what ways are these systems embedded in human interactions, and is it clear to the caller that they are engaging with an AI system, particularly in light of advances in synthetic voice technologies? Of course, those questions also touch upon the question of trust.

In summary, it becomes evident that transparency is both increasingly complex and even more essential. Addressing it requires ethical and technical discourse on the possibilities and limitations of Explainable AI in relation to AI Agents and Agentic AI. In particular, careful consideration is needed regarding *meaningful human control* and the role of the *human in command* (e.g. through logging or monitoring dashboards, participatory design etc.), as oversight mechanisms become harder to maintain in complex multi-agent environments.

### 3.2 Justice and fairness

Another critically important ethical dimension, closely linked to transparency, is justice and fairness [25, 26]. Discussion on potential biases in AI systems and their implications for fairness and discrimination are well established; these concerns are now amplified in the context of AI Agents and Agentic AI.

This amplification arises partly from reduced transparency, but is driven even more by the broad agency of these systems. When AI Agents or Agentic AI systems act technically autonomously, human oversight may be diminished. Consequently, the ability of humans to detect potential biases is even more reduced compared to traditional AI applications. At high levels of automation, users may not even notice if an AI system systematically disadvantages certain marginalized groups while favouring others. For example, an AI Agent used to pre-screen job applications might automatically reject candidates before a recruiter ever reviews them. If the system's selection process is biased, certain applicants may never reach human consideration—leading also to questions of who becomes visible [24]. As a result, the recruiter cannot form a comprehensive picture or review the rejected candidates, since they have been automatically filtered out and notified of their rejection. As the capabilities of the system expand, and its deployment becomes more extensive, addressing fairness becomes increasingly urgent and highlights the ethical importance of explainability of AI systems and a review of the values embedded in these systems.

Fairness risks also extend to users themselves, who may be disadvantaged if their preferences or characteristics do not align with the system's encoded values. AI Agents might implicitly nudge—or even manipulate—users toward certain choices or refuse to execute actions that conflict with their programmed objectives. For instance, consider an AI-based financial assistant that refuses to process a donation to a particular charity because its programmed objective is to maximize savings, or because it implicitly devalues the goals of that organization. This raises the ethical question of who defines the values of AI Agents or Agentic AI systems,

and how these values shape the system's capabilities and limitations.

Another aspect closely related to the question of fairness and justice concerns sustainability and the environmental impact of AI systems [27, 28]. These issues become particularly relevant in the case of multi-agent systems that perform multi-step operations, interact with one another, and call on external tools. Due to their more complex architecture and broader scope of tasks, these systems are likely to consume even more energy, both during training and in operation. In addition, cutting-edge technologies constantly require more advanced hardware, which shortens hardware usage cycles and can contributed to increased electronic waste [29].

Building on this, issues of justice also extend to the question of who is involved at which point within the AI life cycle. This includes, for instance, the exploitation of click workers who contribute to the training of AI systems under poor working conditions—often for minimal pay and without adequate psychological support—despite performing an essential task for AI development. It is crucial to adopt a participatory approach—so that the human remains in command and meaningful human control can be exercised—in order to include a broad range of perspectives in the discourse and to draw attention to inequalities and disadvantages.

Overall, the discussion of fairness and justice illustrates that existing problems of AI systems can be further amplified by the emergence of AI Agents and Agentic AI. The more extensive their agency, influence on human autonomy, and the degree of entanglement between humans and technological systems, the more urgent it becomes to examine and embed value-sensitive design principles despite the fact that a general, global consensus may be hard or impossible to reach [30, 31]. Importantly, this is not a call for a "laissez-faire" approach that permits everything, nor for regulation so restrictive that the system loses practical utility. Rather, it highlights the need for a societal negotiation over which values should guide these systems across the whole life cycle and which types of tasks can appropriately be delegated to AI Agents or Agentic AI.

### 3.3 Non-maleficence

The principle of non-maleficence is closely related to the preceding chapter, as the question of harm intersects with other ethical topics we address in this paper, such as fairness, bias and discrimination, as well as the values embedded in AI systems. Additional questions regarding responsibility or privacy will be discussed in more detail in later subsections. Here, we focus on reflecting more fundamentally on the potential for harm posed by AI Agents and Agentic AI.

These systems—particularly due to their broad agency and advanced capabilities to fulfil a wide range of tasks—have an enhanced capacity to cause harm, thus affecting safety and security. On the one hand, harm can be unintentional, as in cases of bias and discrimination discussed earlier. Misuse or overreliance by users can also cause harm, for example if users overestimate the systems' capabilities or place excessive trust in them. Beyond individual harm, questions arise about the societal implications of the widespread access to personal AI assistants in the form of Agentic AI systems that communicate on behalf of users and manage complex tasks. For example, an appointment assistant might prioritize minimizing wait times, unintentionally rescheduling routine medical appointments for chronically ill patients so far apart that intervals between treatments become medically inappropriate. Overall, the potential of Agentic AI to transform workplaces, administration, and even the judicial system underscores the need for empirical investigation and ongoing ethical reflection. On the other hand, this harm could also be intentional: the systems could be deliberately programmed to cause damage or become infiltrated by a malicious attacker. Their ability to adapt contextually and select among multiple tools, combined with their complexity and lack of transparency further amplifies potential risks, for example in high-stakes and military applications [32]. This, again, shows the importance of classic ethical questions about human oversight and control.

In summary, the classic ethical imperative of non-maleficence is intensified by the emergence of these systems. Their integration into social and organizational workflows and routines as well as their high interconnectivity means the potential for harm is significant, demanding careful oversight and responsible design.

### 3.4 Responsibility and accountability

One of the most important questions regarding the use of AI Agents and Agentic AI concerns responsibility. This issue becomes particularly evident due to the broader differentiation of tasks these systems can perform. When they take on tasks that have a real-world impact, the question of who bears responsibility and who is accountable and liable becomes critical.

Even relatively minor incidents illustrate the challenge. For example, an AI Agent might reserve a table at a restaurant, but the user fails to show up, or an AI Agent could purchase a concert ticket but fails to update the user's calendar due to an integration error, causing the event to be missed. In such cases, determining who is accountable and who bears any resulting costs is not straightforward. This question becomes even more urgent in high-stakes contexts

or situations involving heightened vulnerability, such as in healthcare, finance, or legal settings.

Current research is exploring new frameworks for responsibility and liability in AI systems in general [33]. However, regarding AI Agents and Agentic AI, the question is even further amplified due to their broad agency [16]. As these systems are designed specifically to operate with minimal human intervention, responsibility can become diffuse; augmented by a lack of transparency (including that there is uncertainty about how a machine learning system develops over time) and the complexity of Agentic AI systems. Therefore, a key question concerns how responsibility can be meaningfully attributed in systems with broad agency, and particularly in multi-agent systems, so that no responsibility gaps arise in the event of harm. Since these systems operate with relative autonomy, it becomes essential to determine the point at which responsibility shifts—and how a corridor of action can be established, both legally and technically, within which these systems may operate. One possible way to introduce differentiation would be through levels of automation, analogous to the SAE Levels of Driving Automation (SAE J3016) established in the discussion on autonomous driving, so that, depending on the degree of automation, responsibility is shifted or distributed accordingly [34]. Overall, it is essential that manufacturers are not exempt from their product liability: learning systems cannot be considered fully developed at the time of deployment, but must—due to their dynamic nature—be continuously developed, evaluated, monitored, and improved. This includes, for instance, an obligation on the part of manufacturers to provide updates and, depending on the context, for either the manufacturers or the users to ensure their installation. In this context, prior work on business ethics and corporate social responsibility can offer useful insights [35].

Taken together, it is crucial that AI Agents and Agentic AI—with broad agency, expanded impact on human autonomy, and increased entanglement between humans and technology—are carefully incorporated into these discussions about responsibility and liability. Establishing clear lines, whether for manufacturers, operators, or users, is essential to maintain ethical oversight and responsible governance.

### 3.5 Privacy

Privacy is another critically important ethical concern, arising on at least two levels: (1) the processing and control of data, and (2) the type of data being processed.

The first aspect—data processing—is relevant to Big Data and AI systems in general [36]. On the one hand, data are required for model training; on the other hand, data are processed so that the system can, for example, generate text, compute risk scores, or make recommendations. How these data are processed, where they are stored, and whether they can be deleted is often unclear. In Agentic AI systems, which exchange data across multiple agents, maintaining an overview of where and how data are processed and stored becomes even more opaque. It is also uncertain if, and to what extent, external tools or interfaces process or store these data. In this sense, privacy is closely linked to transparency, as previously discussed. Existing approaches developed in the context of Big Data—such as data sovereignty and data agents [37]—may reach their limits in this context and would require both ethical and technical evaluation.

The second aspect concerns the type of data required. For an AI Agent or Agentic AI system to provide meaningful support, it must have extensive access to personal data (e.g., name, date of birth, address, health data, credit card information, calendar entries, personal chats and emails). This raises important ethical questions: Are highly sensitive data adequately protected? Does the system request user confirmation before sharing personal information? How does it determine which data can be disclosed under which circumstances? For instance, a telephone agent calling a medical practice may need to provide a patient's date of birth to schedule an appointment, but should never share this information with a spam caller. Similarly, the system must assess whether a request, such as processing and storing credit card information, is legitimate (e.g., for a hotel booking) or not. There it becomes evident, how privacy is linked to responsibility and liability, as questions arise about what happens if data are mistakenly shared or accessed by unauthorized parties.

Given the rapid adoption and growing user base of large language models [38], it is likely that the deployment of AI Agents and Agentic AI will continue to expand. Privacy in this context carries significant risks, underscoring the need for clear rules and technical safeguards. Policies should address how data are handled, how users can maintain oversight of their information, and what security measures must be implemented to protect sensitive data and thereby the users (e.g., to respect their autonomy, dignity, integrity and vulnerability) [39].

## 4 Discussion

The review of central ethical principles has demonstrated that classical ethical principles for AI also apply to AI Agents and Agentic AI. However, these principles have become even more critical and urgent, as both the potential benefits and the risks are amplified by these systems. Therefore, the following section reflects on the ethical implications with respect to the three aforementioned characteristics—(1)

broad agency, (2) high impact on human autonomy, and (3) deep entanglement of humans and technology—and sketches initial steps toward a governance approach for AI Agents and Agentic AI.

First, these new systems exhibit broad agency. Since AI Agents and Agentic AI systems are designed to perform complex tasks with minimal to no human intervention, they require more capacity for independent action. To act effectively as assistants, they need almost unrestricted access to personal data and information. However, this already gives rise to potential ethical challenges, for instance concerning justice and fairness, if the system makes biased decisions or causes harm—directly linking to the question of what enhanced agency means for responsibility and liability. To address this increased dynamism, it is necessary to implement technical solutions that embed ethical principles directly in the development and application of AI Agents and Agentic AI, following the concept of *ethics by design* (with the caveat articulated in Subsect. 3.2 concerning value-sensitive design and the aim of achieving the broadest possible consensus on values). Concrete examples for technical solutions include sandboxing approaches, where such systems interact only with test environments and thus cannot cause direct harm, as well as AI Agents that monitor other AI Agents (*guardian agents*) [40]. However, particularly with regard to technical solutions such as guardian agents, the problem of an infinite regress arises: who monitors the guardian agents themselves? Introducing further agents would lead to a paradoxical situation in which an agent monitors an agent that monitors another agent. This illustrates that also within the domain of technical solutions, novel, innovative, and secure approaches are required. Overall, these systems must be technically robust, including safeguards against malicious attacks, regular updates, and ongoing evaluation. The intrinsic dynamism of these systems does not imply that manufacturers can step back; on the contrary, manufacturers are now especially obliged to continuously develop, evaluate, and monitor these dynamic systems. To advance such technical measures, a combination of voluntary industry initiatives and regulatory frameworks is likely necessary. Incentives could encourage companies to implement ethical safeguards proactively, while legal measures ensure baseline standards are maintained. Achieving the right balance between under-regulation and over-regulation will be key to retaining the practical utility of these systems while ensuring their responsible application.

Second, it has been shown that these systems exert a strong influence on human autonomy (understood as the capacity of the embodied human subject to act in a relatively self-determined manner). On the one hand, humans deliberately delegate a certain degree of autonomy to these systems: they shall have relative technical autonomy so that they can perform tasks relatively independently on their behalf. Yet, beyond this voluntary and functional delegation, human autonomy is further restricted, as the opacity of AI systems (especially in the case of Agentic AI) complicates transparency and explainability, thereby making informed decision-making and responsible oversight increasingly difficult. From this perspective, it also becomes necessary to ask whether, and by what means, manipulation by such systems can be identified and prevented. The pressing question remains: how can *meaningful human control* be ensured, so that the *human* remains *in command*? Addressing this requires the participation of diverse stakeholders throughout the entire life cycle of the system. In addition, education and public awareness about the opportunities and risks of AI Agents and Agentic AI are essential. Given their easy accessibility and considerable potential, it can be expected that these systems—similar to LLMs—will experience rapid growth. Responsible use therefore also depends on the knowledge and awareness of users themselves. At minimum, users need a foundational understanding of how these systems operate and what impact they may have on the user's autonomy. Once the operators or users are empowered to exercise control, it is equally important—analogous to the obligations of manufacturers—that they fulfil their responsibilities (e.g., oversight, installing updates, …). In this sense, one could say that the broader the agency of these systems and their impact on human autonomy, the more extensive the required participation and efforts toward human control.

Finally, the entanglement between humans and these systems has been found to be particularly important to consider. Accordingly, there is a need for further ethical, legal, and social (ELSI) research, together with interdisciplinary discourse, along the entire life cycle of AI Agents and Agentic AI. On the one hand, ongoing development processes should be accompanied by risk assessments to identify emerging trends early, evaluate long-term societal and ethical implications, and provide guidance for responsible deployment. On the other hand, a broad, participatory, interdisciplinary dialogue is needed to determine the societal domains in which AI Agents and Agentic AI systems should operate, the purposes for which they are deployed, and the guiding values they should embody. In this way, it is possible to contribute to disentangling the entanglement between humans and AI systems, which arises when AI Agents and Agentic AI systems increasingly take on extensive and complex tasks, some of which are highly personal in nature. Particularly with regard to responsibility and liability, precise differentiation is required (analogous, for example, to the SAE levels) to prevent the diffusion of accountability. Humans must not be placed *out of the loop*; rather, they must retain control. This necessitates considering specific contexts and concrete

use cases, which can be assessed, for instance, through a risk-based approach.

Ultimately, the goal is to harness the significant benefits that AI Agents and Agentic AI offer, while ensuring that these systems are deployed in a responsible and balanced way.

## 5 Limitations

Our paper has three main limitations.

First, our analysis addressed ethical principles at a high level, without engaging in more depth with each principle and its place in the broader ethical discourse. We also limited our scope and clustered the principles, which means that some principles that may be equally relevant and potentially challenged by these novel systems were not addressed here. More broadly, our deductive methodology did not allow us to identify new ethical principles that could emerge from an inductive approach specifically tailored to AI Agents and Agentic AI.

The second limitation concerns the timing of the paper. The development of AI Agents and Agentic AI is still in its early stages, meaning that both the technical capabilities and the ethical reflection surrounding these systems are only beginning to unfold. Therefore, our analysis is necessarily theoretical and could not include any empirical data.

Third, we have no further elaborated on the significance that AI Agents and Agentic AI may have for the transformation of human self-understanding. This also includes questions concerning hybrid agency and the possible development of a posthuman ethics [41]. Future research would need to broaden the scope in this direction, as the more AI Agents and Agentic AI systems are deployed, the more important it becomes to navigate these systems using a compass that is grounded in education (aimed at sensitivity to differences and tolerance of ambiguity, practiced within a community).

Overall, the goal of this paper was not to provide a comprehensive overview, but rather to offer initial perspectives and questions that can serve as an impetus for further ethical research in this rapidly emerging field.

## 6 Conclusion

The scope of this paper was to provide an initial examination of how recent development in AI Agents and Agentic AI relates to established ethical principles. We approach this task on the basis of established ethical principles, proposed by the UNESCO and clustered in a structure based on the most frequently discussed ethical principles according to Jobin et al.: *transparency*, *justice and fairness*, *non-maleficence*, *responsibility and accountability*, and *privacy*.

In summary, our analysis indicates that these principles are still of great relevance for *AI Agents* and *Agentic AI* systems so that the governance of these new systems does not have to start from scratch. However, due to their broad agency, which has a higher impact on human autonomy and leads to an even deeper entanglement between humans and technology compared to traditional AI systems, ethical questions gain even higher importance and urgency.

Considerations for the governance of AI Agents and Agentic AI must necessarily take these characteristics into account. The broader the agency of these systems, the more important is that humans ultimately retain control. This entails technical robustness, broad participation and education of diverse stakeholders, more ELSI research and clear legal frameworks that hold manufacturers accountable, ensuring that the systems are continuously developed, evaluated, monitored and improved throughout their entire life cycle. Questions of responsibility and liability must be clearly delineated; here, a risk-based approach differentiated according to levels of automation could be helpful in preventing the diffusion of responsibility.

With this contribution, we offer a first mapping of how ethical principles are affected and aim to encourage further ethical reflection and research on AI Agents and Agentic AI. These systems hold considerable potential by autonomously managing a wide range of tasks, offering relief in both everyday and professional contexts. At the same time, as we have demonstrated, they raise new and pressing challenges for established ethical principles. Advancing ethical inquiry is therefore imperative to guide the responsible development, deployment and oversight of AI Agents and Agentic AI.

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**Data availability** Data sharing is not applicable to this research as no data were generated or analysed.

## Declarations

**Conflict of interest** The authors declare no competing interests.

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